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The dynamic relationship of knowledge management processes and project success: modeling the mediating role of knowledge worker satisfaction

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Abstract

Purpose – Constructed upon a knowledge-based view, the purpose of this paper aims to empirically examine the entrepreneurial leadership impact as knowledge management enabler on knowledge management processes and knowledge management processes on project success. The study further ascertains the mediating role of knowledge worker satisfaction among knowledge management processes and project success.

Design/methodology/approach – Using the data collected from 302 project workers from the software industry, China. The research used structural equation modeling (SEM) to analyze the hypothesis relationships using smart-PLS 3.2.9.

Findings – The outcome of the study reveals that entrepreneurial leadership has a substantial significant impact on knowledge management processes, and knowledge management processes influence project success via knowledge worker satisfaction as a mediator both directly and indirectly. Moreover, the study found partial mediation of knowledge worker satisfaction between knowledge management processes and project success.

Practical implications – The current research identifies that entrepreneurial leadership may play a role in fostering knowledge management processes in project-based organizations (e.g. software industry) that can use the knowledge management processes to increase their chances of project success. More broadly, the current study contributes to the entrepreneurial leadership, knowledge management processes, knowledge worker satisfaction and project success existing literature and strengthens the relationship and suggest that how project manager's value knowledge worker satisfaction and help organizations gain competitive advantage and project success.

Originality/value – Although there is an increased significance of knowledge management in the software industry, there is a lack of research that examines the enabling factors and outcomes of knowledge management practices. The present study is one of the first studies to ascertain the relationship of entrepreneurial leadership, knowledge management processes, knowledge worker satisfaction and project success. This is one of the initial researches that not only empirically examines the interrelationships among these variables but also enlighten insights into the current literature by instantaneous investigation of the mediating role of knowledge worker satisfaction.

Keywords Entrepreneurial leadership, Knowledge management processes, Knowledge worker satisfaction, Project success, China

Paper type Research paper



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1. Introduction

In a highly competitive, complex, rapidly changing and uncertain situation among organizations, knowledge management becomes a prime concept in academia (Obeidat *et al.*, 2016). Knowledge management emerged in 1990 (Nguyen and Mohamed, 2011) with the recent works in economics and management literature contribute to developing the knowledge-based view. From the perspectives of the knowledge-resource-based view, organizations rely more on their knowledge-based resource (Zack *et al.*, 2009) for superior performance of the organizations (Al-Hakim and Hassan, 2016) and gain competitive advantage (Shujahat *et al.*, 2019). Existing theoretical and empirical literature (Shujahat *et al.*, 2018; Ologbo *et al.*, 2015) adheres to the theory of knowledge-based theory, that is, how firms that manage their knowledge resources tend to perform better than those that do not.

Previous research confirms the organization's capabilities concerning knowledge management processes and enablers (Sahibzada *et al.*, 2020d; Iqbal *et al.*, 2019; Anderson *et al.*, 2006). Various scholars defined knowledge management processes as activities that include knowledge identification, creation, collection, organizing, storage, sharing and application to improve organizational competitiveness and performance (Ali *et al.*, 2018; Sadeghi and Rad, 2018; Obeidat *et al.*, 2016). Meanwhile, knowledge management enablers are the factors, for instance, leadership, culture, incentives and organizational structure, that facilitate the activities related to knowledge management processes (Cho and Korte, 2014; Ho, 2009).

To cope with the emerging business scenario, every organization has a firm aspiration to develop and create competition for competitors at the local, national or global level (Sahibzada *et al.*, 2020d). The goal can be achieved by maximizing and creating new knowledge resources through proven organizational mechanisms (Sahibzada *et al.*, 2020b). Several organizational mechanisms are accustomed to this purpose, but the importance of these is the company's leadership practices (Singh, 2008). Therefore, one of the leading knowledge management enablers is organizational leadership (Iqbal *et al.*, 2019); leaders directly impact organizations and how they should address and manage knowledge management processes (Latif *et al.*, 2020a). Knowledge management is also inevitable for project-based organizations. The information technology (IT) sector, being modern, innovative and knowledge-intensive, is often organized by projects (Handzic *et al.*, 2016). Despite that, the failure rate of the project is around 70%; possible reason for a high percentage of project failure in the organizations is not to utilized the knowledge resources efficiently and fully to increase the project success (Yeong and Lim, 2010; Doherty *et al.*, 2003).

21st century being the "Knowledge era," the provision of knowledgeable business services depends to a large extent on the effective management of knowledge in the organizations (Sahibzada *et al.*, 2020e), and leaders are the central being in the process of managing and effectively utilizing the knowledge resources (Bryant, 2003; Sahibzada *et al.*, 2020c). Also, organizational productivity depends more on managing proper knowledge through a suitable leadership style (Singh, 2008). Nevertheless, conventional leadership cannot be considered as an option only, but entrepreneurial leadership can be identified as an important indicator for actual and successful outcomes for modern organizations (Latif *et al.*, 2020a, b; Abubakar *et al.*, 2018). Entrepreneurial leadership is a blend of entrepreneurship and leadership quality that enables organizations to exploit the opportunities and gain organizational competitiveness (Renko *et al.*, 2015; Huang *et al.*, 2014).

First, earlier studies by Iqbal *et al.* (2019), Sahibzada *et al.* (2019) and Shahzad *et al.* (2016) have testified the impact of knowledge management enablers and knowledge management processes on organizational performance and their effectiveness. However, the theoretical concept is still under-developed (Harrison *et al.*, 2018). The emerging area of entrepreneurial leadership gained particular interest in project-based organizations and performance management (IT sector, Software) because of their relevance to management-based

organizations. But limited research has been done to facilitate knowledge management enabler (entrepreneurial leadership) toward effective implementation of knowledge management processes in the IT sector, particularly in China (Zainol *et al.*, 2018; Mishra and Misra, 2017; Latif *et al.*, 2020b). Therefore, researchers attempt to study the impact of entrepreneurial leadership on knowledge management processes i.e. identification, creation, collection, organizing, storage, sharing and application in the context of IT projects (Naqshbandi and Jasimuddin, 2018; Abubakar *et al.*, 2018; Handzick, 2011; Latif *et al.*, 2020b).

Second, the direct link between the knowledge management enabler and project success may be questionable. Some scholars such as Nasab and Afshari (2019), Chang *et al.* (2018) and Fu *et al.* (2018) assert that knowledge management processes do not directly affect project success; instead, different mediating variables play that role. Thus, the present study has assessed knowledge worker satisfaction as a mediating variable. One stream of research posits that knowledge management could be a way to nurture knowledge worker satisfaction within an organization, enabling the organization to achieve a high level of efficiency (Sahibzada *et al.*, 2020a, b; Shujahat *et al.*, 2018). The role of knowledge worker satisfaction to reinforce the project success has not been explored aptly; thus, there is a need to study the stated variable in project-based organizations (Sahibzada *et al.*, 2020b, e).

Based on the gaps and limitations, the study's objective is to *ascertain the interrelationship between entrepreneurial leadership, knowledge management processes, knowledge worker satisfaction, and project success in the software industry in China.* To bridge limitations, this research assumes to fill the identified knowledge gaps of knowledge management in the software industry and in doing so, makes several theoretical contributions to the existing literature by assessing the interrelationship between entrepreneurial leadership, knowledge management processes, knowledge worker satisfaction and organizational performance. The study first assesses the impact of entrepreneurial leadership on knowledge management processes and further ascertains whether knowledge worker satisfaction mediates the relationship between knowledge management processes and project success. This would help in providing a basis for understanding the mediating mechanism through which knowledge management processes affects the success of projects in the software industry. With significantly limited research on the role of knowledge management in the Chinese software industry, the study would significantly help understand and highlight the role of knowledge management in the Chinese IT industry. The research study endeavors to contributes to the field of knowledge management in the software industry and further understandings towards the two-factor theory of Herzberg (1966) who believed that motivation are intrinsic elements that lead knowledge workers to satisfaction and knowledge-based view (Grant, 1996) literature that stresses knowledge as a critical organizational resource leading to improved organizational performance.

RQ1. Does entrepreneurial leadership has an impact on knowledge management processes in the software industry of China?

RQ2. Is there a direct impact of knowledge management processes on project success in the software industry of China?

RQ3. Does knowledge worker satisfaction mediate the relationship between knowledge management processes and project success in the software industry of China?

2. Hypotheses development and conceptual framework

2.1 Entrepreneurial leadership and knowledge management processes

Effective leadership can be seen as an important tool for sharing related information and knowledge to take timely and practical decisions for organizations (Akram *et al.*, 2019; Chang

et al., 2019). Previous research believes that leadership is a contributing factor to knowledge management (Latif *et al.*, 2020a, b; Iqbal *et al.*, 2019), enabling organizations to adjust knowledge management behavior based on organizational capabilities and strategies optimal strategic communication. Promoting learning organizations through organizational strategies, promoting knowledge management values, and providing indicators that determine the knowledge resources efficiently (Koohang *et al.*, 2017; Wei *et al.*, 2009). Therefore, leadership includes strategic issues and determine how organization define their activities and utilize resources related to knowledge to improve their capabilities (Latif *et al.*, 2020b).

Musa and Fontana (2016) referred to entrepreneurial leadership as a unique leadership style that creates heterogeneous talent and being more creative and innovative in the collective process to address uncertain business environments, creating consistent strategies and innovations. Entrepreneurial leadership is the visionary process for entrepreneurship. Leaders motivate the teams to promote it fast and in an uncertain environment; without their commitments, knowledge management success is difficult to achieve (Bagheri and Pihie, 2009; Civi, 2000).

Leaders play a key role in managing organizational knowledge. They provide vision and motivation to the employees at all levels and promote the knowledge conversion into a competitive advantage through a proper strategy (Bryant, 2003). Prior research also highlighted the importance of leadership in knowledge management literature (Townsend, 2010; Fernald *et al.*, 2005). Studies showed that knowledge management requires at all levels of the organization through the conscious efforts of leaders (Bryant, 2003). Srivastava *et al.* (2006) particularly argued the leadership importance that strengthened sharing of knowledge among followers. The study is related to Bavik *et al.* (2018), who analyzed ethical leadership with knowledge sharing and shows leadership is a crucial element in knowledge sharing. Thus, being recognized as a fundamental aspect, leadership affects interpersonal knowledge sharing in an organization (Kim and Yun, 2015). Moreover, Singh (2008) investigated various leadership styles (management, supportive, consultative, and delegating leadership style) and their impact on an Indian IT industry knowledge management practices.

The knowledge-based view recognizes knowledge as the company's significant strategic asset and resource, and entrepreneurial leadership attempts to discover and use strategic value creation (Gupta *et al.*, 2004; McGrath and MacMillan, 2000). Effective leadership requires proactive entrepreneurial leadership. Leaders must create a culture in the organization to collaborate and share knowledge (Wong and Aspinall, 2004). They also encourage employees to share their ideas with other fellows and creates an atmosphere that welcomes new ideas and strategies (Bryant, 2003). They also provide an environment in which employees create new knowledge and creativity (Surie and Ashley, 2008; Mumford *et al.*, 1997).

In the light of the knowledge-based view, it is assumed that knowledge can be created, stored and utilized not by organizations as a whole but at an individual level. It is a daunting task to coordinate and integrate knowledge held by the individuals (Miles, 2012). Entrepreneurial leadership made it possible, classified by their ability as an integrator and informed (Latif *et al.*, 2020a; Huang *et al.*, 2014). Knowledge can be identified as a pivotal resource and asset to create value in the organization (Zack *et al.*, 2009). According to the knowledge-based theory proposed by Grant (1996), to achieve better performance, utilization and implementation of knowledge-based resources and knowledge management processes are required. A leader has direct control over organizational activities (Bryant, 2003), employees look to an entrepreneurial leadership for a vision, guidance and other tasks (Jansen *et al.*, 2009); these factors also influence worker's motivation to develop innovation, foster creativity and promote knowledge creation and sharing (Chen, 2007).

H1. Entrepreneurial leadership has a significant impact on all of the knowledge management processes (knowledge identification, creation, collection, organizing, storage, dissemination and application).

2.2 Knowledge management processes and project success

Several pioneering studies, especially in IT literature, address defining knowledge by distinguishing knowledge, information and data (Alavi and Leidner, 2001; Feder *et al.*, 2003; King, 2008). Knowledge can be viewed from multiple perspectives: state of mind (fact of knowing) (Schubert *et al.*, 1998), as an object (Zack, 1998), information access condition (McQueen, 1998) or capability (Watson, 2008). These different visions of knowledge lead to different perceptions of knowledge management (Alavi and Leidner, 2001). The dominant perspective is based on a broader vision that considers knowledge management with two main sets of functionalities: infrastructure (Gold *et al.*, 2001) and process capabilities (Dahiyat, 2015; Tseng and lee, 2014). These knowledge management process capacities relate to management procedures and activities, which relate primarily to knowledge resources as inputs that must be properly acquired, developed and translated into useful ways of using them in value-added processes (Ramadan *et al.*, 2017) and contribute to enhanced performance (Leal-Rodriguez *et al.*, 2013).

According to Azan *et al.* (2017), knowledge management processes prime goal is to make an organization aware of its knowledge at the individual and collective levels and use it to shape efficient and effective business processes and competitive advantage. As recent interest in organizational knowledge has prompted knowledge management for the organizational benefits (Teixeira *et al.*, 2018; Gold *et al.*, 2001). Knowledge management processes are described operationally as an intentional process for capturing, creating, storing, transforming, sharing and using the information to serve employees, organizations, and their customers (Jain, 2007). Knowledge management processes capacities formalize knowledge and decision-making and create new competencies to improve organizational performance, profitability through innovation and customer satisfaction (Chiu and Chen, 2016; Shahzad *et al.*, 2016). Although there are extensive studies on knowledge management and its impact on organizational performance (Iqbal *et al.*, 2019; Shahzad *et al.*, 2016), little is known about its impact on project success (Latif *et al.*, 2020b). Yang *et al.* (2017) and Yang *et al.* (2012) have found a significant impact of knowledge management practices on project success. The relationship between knowledge management processes and project success can be further elucidated using a knowledge-based view and found a significant positive outcome (Latif *et al.*, 2020b), while it was moderately significant in Fedor *et al.* (2003).

H2. Knowledge management processes (identification, creation, collection, organizing, storage, dissemination and application) have a significant impact on project success.

2.3 Knowledge management processes, knowledge worker satisfaction and project success

The knowledge-based view emphasizes objective measures, e.g. knowledge-based resources and knowledge management (Chatzoudes *et al.*, 2015), but does not address the subjective measures, e.g. knowledge worker satisfaction (Frank *et al.*, 2015). Knowledge workers input is the knowledge itself; hence, optimizing the provision of knowledge can greatly improve their satisfaction (Kianto *et al.*, 2016). Although, if the outcome of the knowledge management is an organizational competitive advantage, and knowledge management can improve the knowledge worker satisfaction, then through deductive reasoning, Herzberg's theory and past research emphasize that knowledge worker satisfaction leads to better organizational outcomes (Shujahat *et al.*, 2018; Biasutti and Heba, 2012; Singh and Sharma, 2011).

Herzberg's two-factor theory and previous researches help deduce the positive effect of knowledge management processes on knowledge worker satisfaction. Herzberg's two factors theory postulates that there are mutually two exclusive sets of factors. The first one is hygiene factors and the second one is motivators that drive the satisfaction of knowledge workers (Herzberg, 1964, 1966). According to knowledge management mechanisms, knowledge management processes provide the hygiene factor to avoid dissatisfaction and confirm the motivational requirements (Sahibzada *et al.*, 2020a). Knowledge management ensures rewards, capability and assistance in unbiased dealing to avoid dissatisfaction, thereby ensuring hygiene components (Sahibzada *et al.*, 2019; Kulkarni *et al.*, 2006; Bryant, 2003). Additionally, knowledge management processes also impact motivators by facilitating the following motivating factors (Shujahat *et al.*, 2018, 2019; Liu *et al.*, 2017).

(1) Ability to work with the required autonomy.

(2) Creating a knowledge-based cultural environment by emphasizing the knowledge management processes.

(3) Providing the employees with an opportunity to engage with knowledge-related projects to acquire sustainable knowledge that intrinsically motivates them (Nonaka and Takeuchi, 1995).

(4) Obliged the firm personnel to acquire, share and utilize the knowledge.

(5) Finally, make sure that the supporting backend is provisioned.

Many researchers argued that knowledge management is a knowledge worker satisfaction precedent (Kianto *et al.*, 2016; Chatzoudes *et al.*, 2015). These arguments are as follows (Sahibzada *et al.*, 2020a, c; Razmerita *et al.*, 2016).

First, the previous studies suggest that knowledge management should be evaluated by subjective measures, such as the satisfaction of knowledgeable workers rather than objective measures. This theory indicates the importance of knowledge worker satisfaction in measuring the effectiveness of knowledge management (Sahibzada *et al.*, 2020b; Shujahat *et al.*, 2018).

Second, knowledge workers are motivated to solve the problems required for knowledge-related activities rather than undermining the financial incentives (Shujahat *et al.*, 2018). Knowledge management provides employees with the knowledge and problem-solving perspective through a proper, timely strategy and proper place to provide the right knowledge (Sahibzada *et al.*, 2020b, d).

Third, in particular, the current study shows that the creation of knowledge can improve the satisfaction level of knowledgeable employees because the new creation may increase the worker's productivity. Likewise, knowledge exchange ensures that employees' mutual needs are met (Sahibzada *et al.*, 2020a). Finally, the use of knowledge makes work easier. In short, knowledge management processes (knowledge creation, sharing and application) increase the satisfaction and productivity of organizational personnel (Sahibzada *et al.*, 2020b, c).

The relationship between the knowledge management and satisfaction level can be inferred from previous studies (Sahibzada *et al.*, 2020a, b; Butt *et al.*, 2018), which were entitled to knowledge management processes and knowledge worker satisfaction. Many empirical studies related to knowledge management processes, particularly knowledge sharing and satisfaction with the work, incurred both directly or indirectly relationships (Sahibzada *et al.*, 2020a, b; Shujahat *et al.*, 2018; Jadidi *et al.*, 2013), including the interrelation between the knowledge management processes and knowledge worker satisfaction with knowledge management relationship (Sahibzada *et al.*, 2020a). Limited studies are available to check the

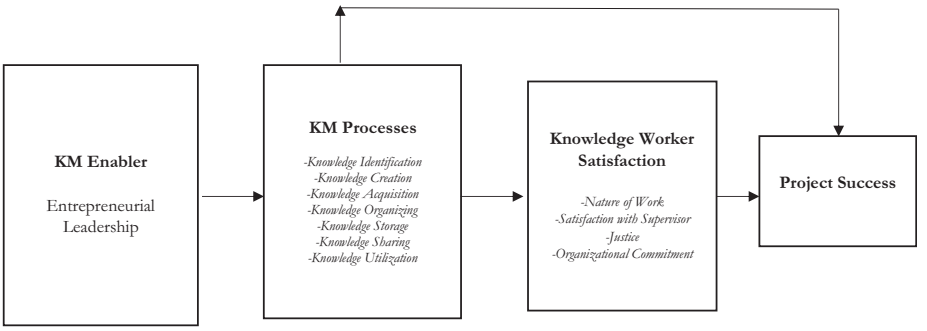
impact of knowledge management processes on knowledge worker satisfaction and project success (Latif *et al.*, 2020a); however, recently, Sahibzada *et al.* (2020b) checked five knowledge management processes, namely, acquisition, utilization, storage and sharing with knowledge worker satisfaction and organizational performance in a knowledge-intensive industry (i.e. higher education). Likewise, a study by Shujahat *et al.* (2018) on the Pakistan IT sector provides contradictory results concerning knowledge management processes, knowledge worker satisfaction and innovation; the result is consistent with the theory of knowledge creation, which adheres to the sole purpose of creating knowledge through innovation.

H3. Knowledge management processes (identification, creation, collection, organizing, storage, dissemination and application) have a significant impact on knowledge worker satisfaction.

H4. Knowledge worker satisfaction has a significant impact on project success.

H5. Knowledge worker satisfaction mediates the relationship between knowledge management processes and project success.

2.4 Conceptual framework



3. Research methodology

3.1 Research firms

Software technology companies make up our study sample because they may have a broader knowledge management strategies spectrum (Donate and Pablo, 2015). Additionally, organizations from high-tech firms need a divergent approach for managing knowledge resources than non-knowledge organizations, and the leadership role must be characterized and supported by the knowledge management processes (Yahya and Goh, 2002).

3.2 Sample and data collection

This research uses triangulation via data source collection (Turner and Turner, 2009), including IT programmers data. Theoretical triangulation is achieved by one or more theoretical frameworks, including the capability models of knowledge management and knowledge-based view (Gold *et al.*, 2001; Grant, 1996). The initial step was to choose the study participants; we included IT respondents (i.e. IT developers). Data for this study were

collected within the time frame from September 2019 to December 2019. Seven hundred questionnaires were distributed totally to three major cities in China (i.e. 373 questionnaires were received, or 53.28 percent). Incomplete and invalid questionnaires were discarded, which was about 71. The remaining 302 questionnaires were utilized for statistics, with a response score of 43.14 percent. This sample size is suitable for examining complex path models using structured equations (SEM) models (Kline, 2011).

3.3 Measures

This study used 56 measurement elements from the existing literature. However, some minor changes have been made to the elements wording to align with the university format (Sahibzada *et al.*, 2019, 2020c). The questionnaire used a five-point Likert scale, ranging from "1" to "strongly disagreed" to "5", as "strongly agree." The eight items scale on entrepreneurial leadership was adapted from Renko *et al.* (2005). The knowledge management processes scale was adapted from Masa'deh *et al.* (2017) with 28 items. The knowledge worker satisfaction scale was adapted from Sahibzada *et al.* (2020a, b, c, d, e) with 14 items. Lastly, six items of project success were adapted from Engelbrecht *et al.* (2017).

3.4 Data analysis procedure

Smart-PLS 3.2.9 package was used for this study (Ringle *et al.*, 2005). PLS-SEM comprises multi-step analysis: the first step is the measurement followed by the second step, i.e. structural model evaluation (Ringle *et al.*, 2018; Wong, 2013). The requirements of the measurement model ensure that only constructs with good indicator load, convergence, compound reliability (CR) and discriminant validity to be used. Evaluation of the structural model is designed to evaluate the path coefficients and test their size and magnitude using a bootstrap method. For mediation evaluation, Preacher and Hayes (2008) technique is the most detailed method of testing the impact of mediation and is best suited to PLS-SEM use (Hair *et al.*, 2014; Hayes, 2009). Recently in knowledge management, studies are conducted by the PLS-SEM Data Assessment Tool (Sahibzada *et al.*, 2020a; Shujahat *et al.*, 2019).

4. Data analysis and results

4.1 Measurement model assessment

The first stage of evaluating the measurement model is to confirm the reliability, dimension and validity of constructs (Hair, 2006). While evaluating the model, none of the elements is ruled out because the factor load is above or near to, with a recommended value of 0.60 for all the factors. Hence, all questions involve the decisive measurement model. Table 1 shows that factor loads of the elements are greater than the (0.60) as recommended value. Besides, CR and AVE of entire concepts are equal to or greater (0.50 and 0.70) than the recommended values, respectively. As a result, convergent validity and reliability are developed. Besides, Table 2 presents the findings for discriminant validity importance (Fornell and Larcker, 1981).

4.2 Structural model assessment

After the measurement model, the second stage was carried out for the evaluation of the structural model. The assumptions are checked sequentially. Firstly, we analyzed the direct impact of entrepreneurial leadership on the knowledge management processes. The second stage analyzed the direct impact of knowledge management processes on knowledge worker satisfaction and project success and the direct impact on project success later. The bootstrap resampling technique was used to establish the significance of the direct path, and 5,000 resamples were utilized (Ringle *et al.*, 2005). Table 3 presents the findings of the hypothesis testing for direct associations. Finally, we tested the impact of the knowledge management

	Λ	α	CR	AVE
Entrepreneurial leadership		0.936	0.947	0.692
EL1	*0.85*			
EL2	*0.81*			
EL3	*0.85*			
EL4	*0.83*			
EL5	*0.86*			
EL6	*0.82*			
EL7	*0.82*			
EL8	*0.82*			
Knowledge management processes				
Knowledge identification		*0.899*	*0.930*	*0.768*
KI1	*0.80*			
KI2	*0.85*			
KI3	*0.81*			
KI4	*0.82*			
Knowledge creation		*0.885*	*0.921*	*0.746*
KC1	*0.81*			
KC2	*0.75*			
KC3	*0.88*			
KC4	*0.86*			
Knowledge acquisition		*0.913*	*0.939*	*0.793*
KA1	*0.82*			
KA2	*0.82*			
KA3	*0.84*			
KA4	*0.88*			
Knowledge organizing		*0.877*	*0.916*	*0.731*
KO1	*0.82*			
KO2	*0.83*			
KO3	*0.79*			
KO4	*0.78*			
Knowledge storage		*0.890*	*0.924*	*0.752*
KST1	*0.77*			
KST2	*0.80*			
KST3	*0.81*			
KST4	*0.87*			
Knowledge sharing		*0.911*	*0.938*	*0.790*
KSH1	*0.81*			
KSH2	*0.81*			
KSH3	*0.81*			
KSH4	*0.82*			
Knowledge utilization		*0.876*	*0.915*	*0.729*
KU1	*0.75*			
KU2	*0.78*			
KU3	*0.81*			
KU4	*0.75*			
Knowledge worker satisfaction				
Nature of work		*0.828*	*0.897*	*0.744*
NOW1	*0.85*			
NOW2	*0.71*			
NOW3	*0.80*			
Satisfaction with supervisor		*0.901*	*0.931*	*0.772*
SWS1	*0.85*			
SWS2	*0.80*			
SWS3	*0.82*			

(*continued*)

Table 1.
Item loadings, reliability and convergent validity

	Λ	α	CR	AVE
SWS4	*0.85*			
Justice		*0.884*	*0.920*	*0.741*
J1	*0.84*			
J2	*0.78*			
J3	*0.810*			
J4	*0.79*			
Organizational commitment		*0.874*	*0.923*	*0.799*
OC1	*0.85*			
OC2	*0.85*			
OC3	*0.81*			
Project success		0.915	0.934	0.702
PS1	*0.84*			
PS2	*0.86*			
PS3	*0.82*			
PS4	*0.81*			
PS5	*0.85*			
PS6	*0.82*			

Table 1.

processes on project success via knowledge worker satisfaction as a mediator. Table 4 presents the assessment results of the mediation.

The results significantly impact entrepreneurial leadership on the knowledge management processes ($\beta = 0.952$, $t = 86.092$, $p < 0.001$) and accepted the H1. Knowledge management processes also significantly positively impacted project success ($\beta = 0.273$, $t = 4.400$, $p < 0.001$), supported our H2. Moreover, the knowledge management processes also significantly impact knowledge worker satisfaction ($\beta = 0.930$, $t = 63.349$, $p < 0.001$); therefore, H3 was also supported. Results also showed significant effects of knowledge worker satisfaction on project success ($\beta = 0.680$, $t = 11.457$, $p < 0.001$). Hence, H1, H2, H3 and H4 are accepted in our study.

4.3 Mediation analysis

Finally, we analyzed the H5 that shows that knowledge worker satisfaction mediates the bond between the knowledge management processes and project success. Both the effects, direct and indirect, showed a significant positive association. Direct effect presents ($\beta = 0.273$, $t = 4.40$, $p < 0.001$), while Indirect effects are also significant at ($\beta = 0.633$, $t = 11.36$, $p < 0.001$), so our results show partial mediation. Suggested that the impact of knowledge management processes on the success of an organization's projects relies on knowledge workers' satisfaction. Therefore, the hypothesis for mediation analysis H5 supported and the findings are shown in Table 4.

5. Discussion and conclusion

5.1 Discussion

An existing study designed to inspects the influence of knowledge management enabler (entrepreneurial leadership) on knowledge management processes and the indirect relationship of knowledge management processes on project success via knowledge worker satisfaction as a mediator in the Chinese IT sector.

For this purpose, a different hypothesis was proposed and was tested using the PLS-SEM technique. The study found a significant positive impact of entrepreneurial leadership on project success and a significant impact of different knowledge management processes on

	EL	J	KA	KC	KI	KO	KSH	KST	KU	NOW	OC	PS	SWS
EL	*0.832*												
J	0.816	*0.861*											
KA	0.807	0.827	*0.890*										
KC	0.821	0.833	0.802	*0.864*									
KI	0.811	0.815	0.889	0.809	*0.876*								
KO	0.825	0.818	0.814	0.814	0.810	*0.855*							
KSH	0.813	0.777	0.820	0.859	0.806	0.818	*0.889*						
KST	0.819	0.847	0.877	0.875	0.854	0.876	0.853	*0.867*					
KU	0.805	0.824	0.809	0.839	0.797	0.818	0.876	0.849	*0.854*				
NOW	0.814	0.799	0.810	0.832	0.800	0.807	0.801	0.811	0.843	*0.863*			
OC	0.822	0.868	0.811	0.844	0.828	0.799	0.801	0.830	0.817	0.823	*0.894*		
PS	0.817	0.894	0.856	0.854	0.826	0.839	0.837	0.861	0.877	0.855	0.887	*0.838*	
SWS	0.807	0.837	0.829	0.828	0.819	0.833	0.809	0.838	0.831	0.858	0.859	0.823	*0.878*

Note(s): The Data on the diagonal (in Italics) is the square root of AVE of the construct while the other values are the correlations with other constructs. *$p < 0.001$

Table 2.
Discriminant validity (Fornell and Larcker criterion)

various dimensions of knowledge worker satisfaction in the IT sector in China, an emerging economy with an unexplored geographical context in the field of knowledge management (Sahibzada *et al.*, 2020b) and under-developed concept in entrepreneurial leadership (Latif *et al.*, 2020b). Outcomes of this study are parallel to the findings of Zaman *et al.* (2019) and Abubakar *et al.* (2018), which shows that entrepreneurship is a recognized path to the growth and strategic renewal in the organizations based on projects (Hitt *et al.*, 2002). This shows that organizations with entrepreneurial characteristics, e.g. research, collaboration, team building and knowledge sharing having a high propensity for a project to be successful (Renko *et al.*, 2015). The result further strengthens the fact that leaders characterized by entrepreneurial leadership competencies tend to deal with a modern organization with a turbulent competitive environment (Luu, 2017; Gupta *et al.*, 2004).

Furthermore, the study ascertained the significant positive impact of entrepreneurial leadership on seven knowledge management processes, and findings validated the hypothesis. Knowledge management plays a crucial role in sustaining a competitive advantage, organizational performance, and project success (Jaleel *et al.*, 2019; Solomon, 2001). Successful knowledge management requires proactive entrepreneurial leadership (Wong and Aspinwall, 2004). In today's business environment, adequate knowledge management is necessary to implement a project (Polyaninova, 2011) successfully. Leadership behavior can be seen as an essential factor as leaders directly impact the direction and effectiveness of knowledge management within the particular organization (Latif *et al.*, 2020b; Nguyen and Mohamed, 2011). From this perspective, entrepreneurial leadership behaves like a dynamic capability with the focus to reconfigure the firm's knowledge management processes and articulating, codifying and utilizing the tacit and explicit knowledge inside the organization (Donate and Pablo, 2015; Zollo and Winter, 2002). The result affirms that the employees' strong entrepreneurial leadership position obliges the company to embark on development initiatives and substantial investments relevant to knowledge management to impact organizational performance and project success (Donate and Pablo, 2015; King, 2009). On the other hand, poor knowledge management and entrepreneurial leadership position adversely impact the project success (Love *et al.*, 2003).

Finally, the results of the current study knowledge worker satisfaction as a mediator have a significant impact on knowledge management processes and project success. Results deliver significant practical assessment and indicate that knowledge management processes positively influences knowledge worker satisfaction, which shows that knowledge

Table 3.
Results of structural model path coefficient (direct relationships)

Hypotheses	Relationship	β	SD	t-value	p-value	Decision
H1	EL → KMP's	0.952	0.011	86.092	0.000	Accepted
H2	KMP's → PS	0.273	0.063	4.400	0.000	Accepted
H3	KMP's → KWS	0.930	0.015	63.349	0.000	Accepted
H4	KWS → PS	0.680	0.059	11.457	0.000	Accepted

Table 4.
Summary of mediation results

Total effect (KMPs → PS)		Direct effect (KMPs → PS)		Indirect effects of KMPs on PS				
Coefficient	p-value	Coefficient	p-value		Coefficient	SD	t-value	p-values
0.905	0.000	0.273	0.000	H5: KMPs → KWS → PS	0.633	0.056	11.36	0.000

management processes can help the Chinese IT sector increase the level of satisfaction of their employees, which ultimately boosts the performance level. Results validated this hypothesis and are consistent with the findings of the previous studies (Sahibzada *et al.*, 2020a, b; Shujahat *et al.*, 2018) related to knowledge management and knowledge worker satisfaction.

The current study indicates that knowledge management processes improve knowledge worker satisfaction and organizational productivity by improving knowledge systems, and improved organizational practices, teamwork, skills and cognitive abilities among employees and significantly improved the chances of project's success. These results validate the knowledge-based theory (Gold *et al.*, 2001; Grant, 1996). On the other hand, a study by Shujahat *et al.* (2018) assessed the impact of knowledge management processes (identification, creation, collection, organizing, storage, dissemination and application) on knowledge worker satisfaction in the IT sector of Pakistan and a contradictory result was found, that knowledge sharing with a significant while knowledge utilization showed insignificant impact on knowledge worker satisfaction. However, the former investigations support the notion that knowledge management processes implicitly connected to the project success by fostering worker satisfaction (Iqbal *et al.*, 2019; Shujahat *et al.*, 2018). Furthermore, this study validates the Herzberg theory and argues that knowledge management processes expedite knowledge worker satisfaction (Sahibzada *et al.*, 2020e; Herzberg, 1974).

5.2 Conclusion

The current study adds to the knowledge management literature focusing on promoting the knowledge management processes (identification, creation, acquisition, organizing, storage, utilization) through knowledge worker satisfaction (job nature, supervisor satisfaction level, justice and organizational engagement), and project success in the IT sector of China. Data were analyzed using the PLS-SEM technique, and findings support that entrepreneurial leadership can significantly improve knowledge management processes, which ultimately increases a project's success chances. The study also ascertains the mediating role of knowledge worker satisfaction and found a substantial significant relationship between knowledge management processes and project success of the IT industry.

5.2.1 Implications of the study. Based on the research results, it is recommended that the decision-makers and management of these organizations formulate contemplate strategies to upskill themselves. It focuses on improving and ensuring development, providing personnel with work support, communication, rewards and incentives, ultimately increasing motivation and mutual relationships among workers. Moreover, it rouses, shapes and maintains employee quality assurance and knowledge management processes (Sahibzada *et al.*, 2019, 2020b). Similarly, administrators should practice well proportionate structures to create, acquire, store, exchange and utilize the knowledge that can lead to the growth of knowledge worker satisfaction, which ultimately leads to input of ideas and innovative outcomes (Sahibzada *et al.*, 2020b). Moreover, the research is groundbreaking on its theoretical basis and attempts to fill the knowledge gap and provides further insights into the knowledge-based view literature by explaining how the knowledge management processes can influence the performance through knowledge worker satisfaction mediation (Sahibzada *et al.*, 2020b). The study empirically confirms the role of entrepreneurial leadership in enhancing knowledge management processes leading to the project's success and further highlights the uniqueness of entrepreneurial leadership compared to other leadership styles (Latif *et al.*, 2020b). The results also suggest that project management practices may be consistent with the firm's leadership, enabling it to achieve better results and a competitive advantage in its project. The project manager must focus on improving entrepreneurial leadership capabilities and improving the organization's knowledge management processes that impact the project's success. The findings of this study indicate that the identification,

creation, acquisition, organizing, storage, exchange and use of knowledge in the software industry can increase the potential of knowledge worker satisfaction and thereby improve project success.

5.2.2 Limitations and future research directions. The limitations of this research opened novel prospects, multi-group analysis with a contrast of cross-cultural study may authenticate the knowledge management effectivity. Instead of entrepreneurial leadership, a knowledge management enabler examined in this study, sustainable leadership and web 2.0 should be studied as knowledge management enablers in IT industry or any knowledge-intensive industry. Furthermore, considering the vagueness of indirect relationship among knowledge management processes and project success, green intellectual capital, green innovation, knowledge quality and knowledge worker performance should be focused on in further studies. At last, big data-based knowledge management should also be considered instead of knowledge management processes with project success or organizational performance in forthcoming studies in knowledge-intensive organizations.

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